

Physical Chemistry

Kinetics Of Bimolecular Reactions

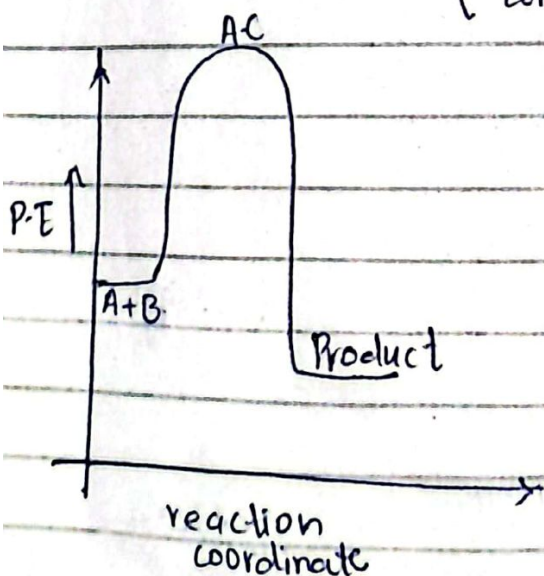
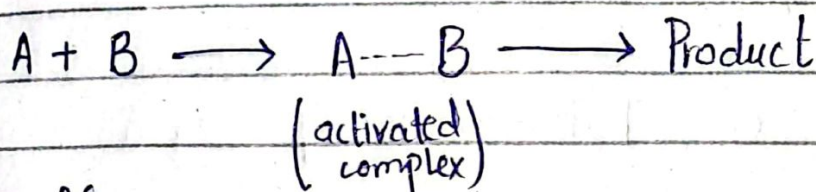
Topic : Effect Of Temperature on the rate of reaction

The rate constant (k) of a reaction is a function of temperature. Any factor affecting the rate constant also affects the rate of reaction. The rate of reaction is therefore also dependent upon temperature.

Conditions for chemical reaction

- Reactants must collide
- collide with proper energy
- collide with proper orientation

Consider a reaction



Activation energy:

Minimum amount of energy, greater than average energy required by reactants to form product is called activation energy.

As the temperature of the reaction mixture increases, the kinetic energy of the molecules also increases. Due to high kinetic energy, there will be greater number of effective collisions and as a result the reaction will take place at a faster rate.

→ Temp $\propto$ K.E $\propto$ no. of effective collisions $\propto$ Rate of reaction.

According to Arrhenius equation.

$$k = Ae^{-E_a/RT} \quad \text{--- (i)}$$

here

k = rate constant

A = Arrhenius constant or pre exponential factor

R = General gas constant

T = Temperature

E_a = Activation Energy

Taking $\ln$ on both sides of eq (i)

$$\ln k = \ln (Ae^{-E_a/RT})$$

$$\ln k = \ln A + \ln e^{-E_a/RT}$$

$$\ln k = \ln A - \frac{E_a}{RT} \ln e$$

$$\boxed{\ln k = \ln A - \frac{E_a}{RT}}$$

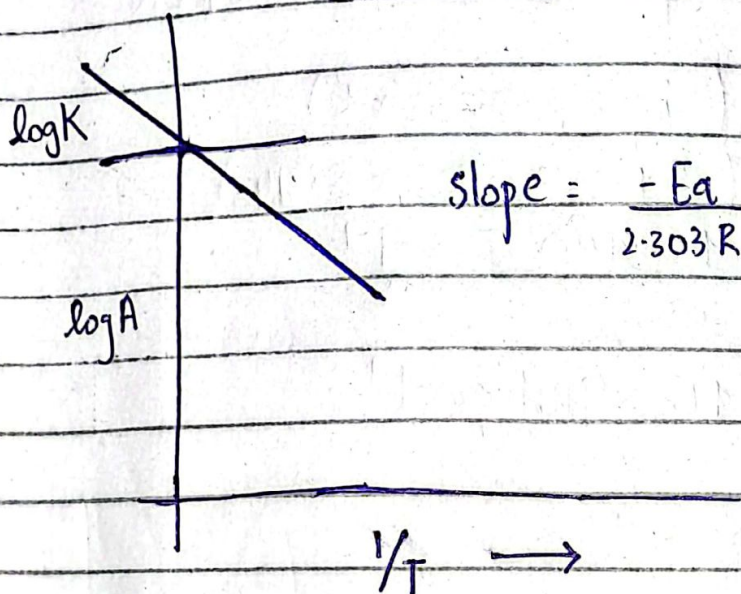
$$\ln K = \frac{-E_a}{RT} + \ln A$$

$$\ln K = \frac{-E_a}{R} \cdot \frac{1}{T} + \ln A$$

$$\frac{2.303 \log K}{2.303} = \frac{-E_a}{2.303 R} \cdot \frac{1}{T} + \frac{2.303 \log A}{2.303}$$

$$\log K = \frac{-E_a}{2.303 R} \cdot \frac{1}{T} + \log A$$

Plotting graph from this equation.



$$\text{slope} = -E_a / 2.303 R$$

$$E_a = -(\text{slope} \times 2.303 \times R)$$

$$\text{Intercept} = \log A$$

$$A = \text{antilog}(\text{intercept})$$

Conclusions from Arrhenius Equation :

- On increasing the temperature, the rate constant also increases until it becomes equal to the pre-exponential factor, A , as the temperature approaches infinity.
- If the value of activation energy is positive then k , increases with temperature.
- If the activation energy is negative, k will decrease with increase in temperature.

Example :

N_2O_5 (Decomposition) slope = -5400K

$$E_a = -(\text{slope} \times 2.303 \times R)$$

$$E_a = -(-5400 \times 2.303 \times 8.314\text{J mol}^{-1}\text{K}^{-1})$$

$$E_a = 103410\text{ J mol}^{-1}$$

or

$$E_a = 103.410\text{ KJ mol}^{-1}$$

Evaluation Of Energy of

Activation between two temperatures

From Arrhenius equation ,

$$\ln k = \ln A - \frac{E_a}{RT}$$

differentiate this equation w.r.t Temp.

$$\frac{d \ln k}{dT} = \frac{d}{dT} \left[\ln A - \frac{E_a}{RT} \right]$$

$$\frac{d \ln k}{dT} = 0 - \frac{E_a}{R} \cdot \frac{d}{dT} \left(\frac{1}{T} \right)$$

$$\frac{d \ln k}{dT} = \frac{-E_a \cdot d T^{-1}}{R \cdot \frac{dT}{dT}}$$

$$\frac{d(\ln k)}{dT} = \frac{-E_a \left(\frac{-1}{T^2} \right)}{R}$$

$$\frac{d \ln k}{dT} = \frac{+ E_a}{RT^2}$$

$$\int_{k_1}^{k_2} d \ln k = \frac{E_a}{R} \int_{T_1}^{T_2} T^{-2} dT$$

$$\ln k \Big|_{k_1}^{k_2} = \frac{E_a}{R} \left[\frac{-1}{T} \right]_{T_1}^{T_2}$$

$$\ln K_2 - \ln K_1 = \frac{E_a}{R} \left[\left(\frac{-1}{T_2} \right) - \left(\frac{-1}{T_1} \right) \right]$$

$$\ln \left(\frac{K_2}{K_1} \right) = \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\ln \left(\frac{K_2}{K_1} \right) = \frac{E_a}{R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Transition State Theory

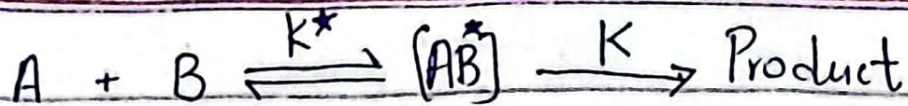
This theory was formulated in 1935 by Henry Eyring.

Postulates

Following are the postulates of Transition state theory.

1. The reaction between two species A and B leads to the formation of an activated complex.
2. Activated complex is highly unstable, energetic specie that spontaneously decomposes to give products.
3. The reaction between the reactants and the activated complex is reversible.
4. Activated complex is in dynamic equilibrium with reactants.
5. Formation of products from activated complex is an irreversible reaction.
6. Rate of formation of product depends upon the concentration of activated complex.
7. The concentration of the activated complex is not a measurable quantity.
8. Rate of reaction is a function of vibrational frequency of activated complex.

From these postulates, the reaction between two species A and B can be represented as



K^* = equilibrium constant

K = rate constant

Eyring Equation :

$$K = \frac{RT}{N_A h} K^* \quad \text{--- (i)}$$

$$K_z = \text{or} \quad \frac{K_B T}{h} e^{-\Delta G^*/RT} \quad \text{--- (ii)}$$

Relation from thermodynamics

$$\Delta G^* = -RT \ln K^*$$

$$\ln K^* = \frac{-\Delta G^*}{RT}$$

$$K^* = e^{-\frac{\Delta G^*}{RT}}$$

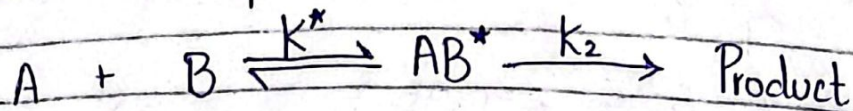
put the value of K^* in eq (i)

$$K = \frac{RT}{N_A h} e^{-\frac{\Delta G^*}{RT}}$$

This equation shows, rate of reaction (rate constant K) depends only on Gibbs free energy, as all other factors are constant here.

Derivation Of Eyring Equation

From the postulates of Transition state theory the reaction between two species A and B can be represented as ;



$$K^* = \frac{[AB^*]}{[A][B]}$$

$$K^*[A][B] = [AB^*] \text{ ——— (i)}$$

Rate of formation of products

$$\frac{dP}{dt} \propto [AB^*]$$

$$\frac{dP}{dt} = \cancel{K^*[A][B]} K^*[AB^*]$$

K^* indicates vibrational frequency in this equation.

Acc. to statistical thermodynamics

$$K^* = \frac{K_B T}{h}$$

K_B = Boltzmann constant

h = Planck's constant

T = Temperature

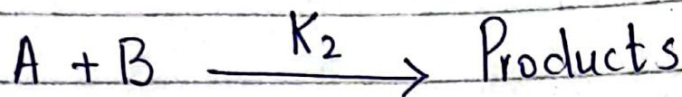
Putting value of K^* in above equation

$$\frac{dP}{dt} = \frac{k_B T}{h} [AB^*] \quad \text{--- (ii)}$$

put the value of $[AB^*]$ in eq (ii)
from eq (i)

$$\frac{dP}{dt} = \frac{k_B T}{h} K^* [A][B] \quad \text{--- (iii)}$$

Now, consider a simple bimolecular reaction



$$\frac{dP}{dt} \propto [A][B]$$

$$\frac{dP}{dt} = K_2 [A][B] \quad \text{--- (iv)}$$

By comparing eq iii and iv

$$K_2 [A][B] = \frac{k_B T}{h} K^* [A][B]$$

$$K_2 = \frac{k_B T}{h} K^* \quad \text{--- (v)}$$

According to Gibb's Free energy

$$\Delta G^* = -RT \ln K^*$$

$$\ln K^* = -\frac{\Delta G^*}{RT}$$

taking antilog

$$K^* = e^{-\Delta G^*/RT}$$

Put this value of K^* in eq ②

$$K_2 = \frac{K_B T}{h} e^{-\frac{\Delta G^*}{RT}}$$

Another form of Eyring equation :

By the definition of Gibbs Free energy

$$\Delta G^* = \Delta H^* - T \Delta S^*$$

put value of ΔG^* in above equation

$$K_2 = \frac{K_B T}{h} e^{-\frac{(\Delta H^* - T \Delta S^*)}{RT}}$$

$$K_2 = \frac{K_B T}{h} e^{-\Delta H^*/RT} \cdot e^{\frac{T \Delta S^*}{RT}}$$

$$K_2 = \frac{K_B T}{h} \cdot e^{-\Delta H^*/RT} \cdot e^{\Delta S^*/R}$$

Collision Theory Unimolecular

Lindmann's Mechanism :

Reactants must collide with proper energy and orientation to convert into product.

Only one molecule is involved.

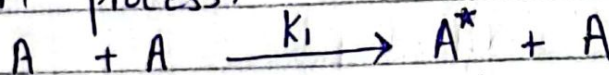


$$-\frac{d[A]}{dt} \propto [A]$$

$$-\frac{d[A]}{dt} = k[A] \quad \text{--- (i)}$$

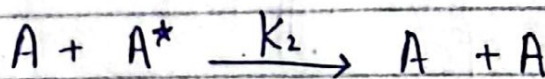
i- Activation process :

1st step of reaction is activation process :



$$\text{Rate} = k_1[A]^2 \quad \text{--- (ii)}$$

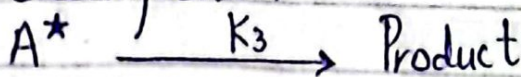
ii- Deactivation :



$$\text{Rate} = k_2[A][A^*] \quad \text{--- (iii)}$$

iii- Decomposition :

Activated molecules may undergoes decomposition.



$$\text{Rate} = k_3[A^*] \quad \text{--- (iv)}$$

Calculating concentration of A^*

from eq (ii)
Rate of appearance of $A^* = K_1[A]^2$

from eq (iii) and (iv)

Rate of disappearance of $A^* = K_2[A^*][A] + K_3[A^*]$

Applying steady state approximation,

Rate of appearance of $A^* =$ Rate of disappearance of A^*

$$K_1[A]^2 = K_2[A^*][A] + K_3[A^*]$$

$$K_1[A]^2 = [A^*]\{K_2[A] + K_3\}$$

$$[A^*] = \frac{K_1[A]^2}{K_2[A] + K_3} \quad \text{--- (5)}$$

putting value of $[A^*]$ in eq (4)

$$-\frac{d[A]}{dt} \text{ (rate)} = K_3 \frac{K_1[A]^2}{K_2[A] + K_3}$$

$$\text{Rate of reaction} = K_3 \frac{K_1[A]^2}{K_2[A] + K_3} \quad \text{--- (6)}$$

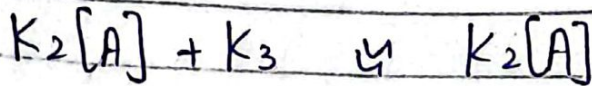
Now, let us consider this eq expression at different pressures.

i) At high Pressure

$$k_2 \gggg k_3$$

At high pressure, the chances of deactivation are more than the chances of decomposition

So, in eq (6)



so eq (6) becomes

$$\text{Rate} = \frac{k_3 k_1 [A]^2}{k_2 [A]}$$

$$\text{Rate} = \frac{k_3 k_1 [A]}{k_2}$$

At high pressure, the reaction follows 1st order kinetics.

ii) At low Pressure

$$k_3 \gggg k_2$$

chances of deactivation are less while chances of decomposition are more at low pressure.

so; eq 4 can be written as

$$k_2[A] + k_3 \rightleftharpoons k_3$$

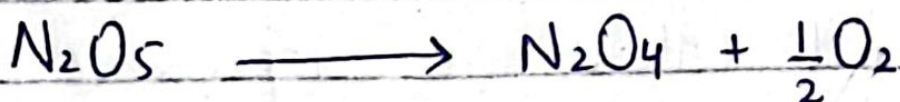
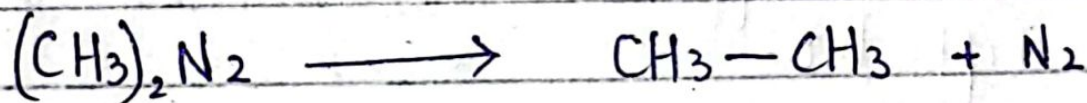
so,

$$\text{Rate} = \frac{k_3}{k_3} \frac{k_1[A]^2}{k_3}$$

$$\boxed{\text{Rate} = k_1[A]^2}$$

At low pressure, reaction follows 2nd order kinetics.

Examples of Unimolecular reaction



Collision Theory Bimolecular

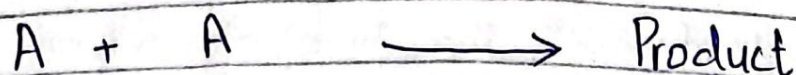
Bimolecular collision theory, for reactions consisting of two reactant molecules only, was proposed by Max Trautz and William Lewis.

Postulates :

- i- Reaction between two molecules takes place due to collision between them.
- ii- All collisions between the molecules do not lead to a chemical reaction.
- iii- The collisions which causes chemical reaction are called effective collisions, the other being ineffective.
- iv- For the collisions to be effective, reactant molecules must collide with proper energy and orientation.
- v- Rate of reaction $\propto$ no. of collisions per unit volume per unit time.
- vi- Rate of reaction $\propto$ fraction of molecules with proper orientation and energy.
- vii- Variation in the rate of reaction due to change in temperature and pressure is due to the fact, that these factors change the probability of the collisions between reactant molecules.
- viii- A catalyst increases rate of reaction by lowering activation energy, hence increasing the probability of effective collisions.

Derivation

Let us write a simple bimolecular reaction for two same molecules.



$$\text{Rate} \propto [A]^2$$

$$\text{Rate} = k[A]^2 \quad \text{--- (i)}$$

$$\text{Rate} = k_n A^2$$

if, concentration of A is 1 mole per dm^3 then

$$\text{Rate} = k \times 1$$

$$\text{Rate} = k$$

According to collision theory, rate of reaction is given by

$$\text{Rate of reaction} = Z_{11} \times q \quad \text{--- (a)}$$

Z = no. of collisions between reacting molecules per dm^3 per second.

11 = it shows two molecules are same.

q = Fraction of molecules which are in activated state (having high energy).

$$\text{No. of collisions : } Z_{11} = \frac{1}{2} \sqrt{2} \pi \sigma^2 \bar{u} n_A^2 \quad \text{--- (ii)}$$

here

σ = collision diameter

$$\bar{u} = \text{Average velocity} = \sqrt{\frac{8RT}{\pi M}}$$



∴ Collision diameter (σ) is the closest distance between centers of two molecules at the time of collision.

Putting value of $\bar{u}$ in eq (ii)

$$Z_{11} = \frac{1}{2} \sqrt{2} \pi \sigma^2 n_A^2 \sqrt{\frac{8RT}{\pi M}}$$

Simplifying

$$Z_{11} = 2\sigma^2 n_A^2 \sqrt{\frac{RT}{M}}$$

$$Z_{11} = 2\sigma^2 n_A^2 \sqrt{\frac{\pi RT}{M}} \quad \text{--- (3)}$$

$$q = \frac{n^*}{n} = e^{-E_a/RT}$$

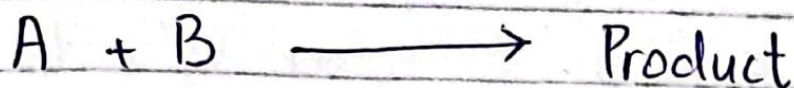
put this in eq (a)

$$\text{Rate of reaction} = 2\sigma^2 n_A^2 \sqrt{\frac{\pi RT}{M}} \cdot e^{-E_a/RT}$$

Now,

For the two different molecules reacting to give product

let



$$Z_{12} = \sigma_{A-B}^2 n_A n_B \sqrt{\frac{8\pi RT}{M}}$$

here M is reduced mass

$$M = \frac{M_A M_B}{M_A + M_B}$$

$$\text{Rate of reaction} = \sigma_{A-B}^2 n_A n_B \sqrt{\frac{8\pi RT}{M}} \cdot e^{-E_a/RT}$$

Limitations :

- Valid only for gaseous phase reactions.
- Steric factors, unexplained.
- Rate of fast reactions, unexplained.
- Reaction at the surface of the catalyst, unexplained.

Comparison of Collision Theory and Transition state Theory

Collision Theory	Transition state Theory
Statement	

For a reaction to occur, reacting molecules must collide with proper energy (sufficient energy) and with the correct orientation. Only a fraction of these collisions, known as effective collisions, result in the formation of products. The rate of reaction is directly proportional to the no. of successful collisions per unit time.

During a chemical reaction, reactants pass through a high-energy, unstable intermediate state called the transition state (or activated complex). The transition state represents the highest energy point along the reaction path between the reactants and the transition state, and the reaction proceeds when the transition state breaks down to form products.

Reaction Rate

The rate of reaction is proportional to the frequency of effective collisions. The rate constant is influenced by the frequency of collisions, activation energy and temperature.

The reaction rate is expressed in terms of equilibrium between reactants and the transition state. It also involves the transmission coefficient (probability that the transition state will proceed to form products).

Mathematical Expressions

Reaction rate is expressed using Arrhenius equation:

$$k = A e^{-E_a/RT}$$

A = Frequency factor
 E_a = Activation energy
 R = gas constant
 T = Temperature.

Rate constant is expressed as

$$k = \frac{k_B T}{h} e^{-\frac{\Delta G^\ddagger}{RT}}$$

or

$$k = \frac{RT}{N_A h} e^{-\frac{\Delta G^\ddagger}{RT}}$$

T = temp, h = plank's constant
 $\Delta G^\ddagger$ = Gibbs free energy.

Applications

Gas Phase reactions

Works well for explaining simple gas-phase reactions, where molecules behave like hard spheres.

Applicable to both gas-phase and solution reactions, including more complex systems.

Temp. effect explanation

Explains how increasing temperature boosts reaction rates due to more frequent and energetic collisions.

Provide a more detailed explanation of temperature effect by incorporating thermodynamic quantities.

Complex reactions

Limited to simple reactions and struggles with complex, multi-step processes.

Suitable for analyzing complex reaction mechanisms, including those with multiple intermediates and steps.

Limitations .

on the
basis of
free energy

Applicable to only simple gaseous-phase reactions.

Does not account for complex reactions, intermediate species or the role of solvents.

Rate of fast reactions, unexplained.

Unable to explain reactions at the surface of catalyst

Incorporates thermodynamic parameters but assumptions like quasi-equilibrium may not apply universally!

More computationally complex, especially when determining transition state energies and structures.